

Enterprise Learning Platform

with Skill and Career Guidance System

Milestone 4 — Project Documentation

AI Intelligence Layer, Instructor & Admin Tools, and Production Readiness

Team C

Prepared for team review

1. Overview — Where We Resumed From

By the end of Milestone 3, the platform had grown from an authentication skeleton into a full student-facing learning and career-guidance product — courses, lessons, quizzes, assignments, certificates, a resume builder, and a community layer. Milestone 4 completes the platform in three directions: adding the AI-powered guidance features that give the platform its “Skill and Career Guidance” intelligence, building out Instructor and Admin tooling so the platform can be run and moderated by more than just students, and hardening the system for production use.

2. Milestone 4 — Feature Summary

2.1 AI-Powered Guidance

- AI Chatbot — a general question-answering assistant available throughout the platform, giving students on-demand help without waiting on an instructor.
- AI Roadmap — generates a personalized learning path for a student based on their current progress and goals, turning the platform from a static catalog into an adaptive guide.
- Skill Gap Analysis — compares a student's current skills against a target role or path and highlights specifically what to focus on next.

All three AI features are served through a single backend `AiAssistantController`, which centralizes prompt construction and calls out to the platform's configured AI provider — keeping API keys and prompt logic server-side rather than exposed to the browser.

2.2 Instructor Tools

- Instructor Dashboard — a role-specific home page summarizing an instructor's own courses and student activity.
- Manage students enrolled in a course — view roster and progress per course.
- Manage a course's lessons — instructors can maintain lesson content for the courses they own, rather than relying on developer-only seeding.

2.3 Admin Tools

An Admin Dashboard gives platform administrators a single view of overall health: total users, active users, the student-vs-instructor split, total courses, total enrollments, revenue, and top-performing courses — the operational layer needed to run the platform as a real product rather than a single-user demo.

2.4 Role-Based Access Control

With three distinct user types now active on the platform — student, instructor, and admin — access control was finalized so each role only reaches the pages and API endpoints meant for it, enforced consistently on the backend rather than only hidden in the frontend UI.

2.5 Background & System Features

- `ReminderScheduler` — a scheduled job that periodically checks for inactive learners and automatically sends reminder emails.

- DataSeeder — populates the database with sample users, courses, and activity on first run, so the platform is demo-ready immediately after setup.
- Standardized API responses (ApiResponse, ApiErrorResponse, PageResponse) and a global exception handler, so every endpoint across all four milestones returns data and errors in a predictable, consistent shape.

3. Backend Structure — Full Picture

By this milestone, the backend follows a clean layered architecture across all feature areas: Controller (handles HTTP requests) → Service (business logic) → Repository (database access) → Model/Entity (data structure), with DTOs shaping API responses. The controllers introduced in this milestone are:

Controller	Responsibility
AiAssistantController	AI chatbot, roadmap generation, and skill-gap analysis
InstructorController	Instructor-side course & student management
AdminController	Platform-wide admin statistics

4. Production Readiness Checklist

The following items were identified across earlier milestones as things to resolve before any public/production deployment. This milestone is where they are formally tracked and, where completed, verified:

Item	Status
Remove or lock down the dev-only demo-data seeding endpoint	Tracked — action required before launch
Rotate Google OAuth client secret, JWT secret, and mail credentials	Tracked — action required before launch
Cloud database migration (from local MySQL)	Tracked — action required before launch
Confirm all AI features are wired to a real/production AI service, not placeholder logic	Verified in this milestone
Role-based access control enforced on the backend for all three roles	Verified in this milestone
Rate limiting on login / forgot-password (brute-force protection)	Tracked — recommended before launch
Backend input validation on all public-facing forms	Tracked — recommended before launch
Merge feature branches into main via reviewed Pull Requests	In progress

5. Current Project Status

Area	Status
AI Chatbot	Complete & verified
AI Roadmap generator	Complete & verified
Skill Gap analysis	Complete & verified
Instructor Dashboard & course/student management	Complete & verified
Admin Dashboard & platform statistics	Complete & verified
Role-based access control (student / instructor / admin)	Complete & verified
Reminder scheduler & data seeding	Complete & verified
Standardized API responses & global error handling	Complete & verified
Full production hardening (secret rotation, cloud DB, rate limiting)	Pending — see Section 4

6. Summary — End-to-End Platform

Across all four milestones, the Enterprise Learning Platform with Skill and Career Guidance System has grown from a login-only skeleton into a complete product: secure authentication, a real dashboard, a full course and lesson experience with graded assessment, assignments and practice, certificates and verification, a resume builder, internships and live sessions, a community forum and leaderboard, AI-powered personalized guidance, and dedicated tooling for instructors and administrators — all backed by a consistent, layered Spring Boot API and a MySQL database.

7. Recommended Next Steps

- Complete the remaining production-readiness items in Section 4 before any public launch.
- Add automated tests (unit and integration) across controllers and services to protect against regressions as the codebase has grown across four milestones.
- Gather real user feedback on the AI roadmap and skill-gap features specifically, since their usefulness depends heavily on prompt quality and real usage patterns rather than code correctness alone.
- Plan a formal versioned release once the feature branch is merged into main via reviewed Pull Request.